setup can be done just once, after which the data can be exported and then imported on other computers. See Chapter 9: Tools Menu for more information about the *Database Export* and *Database Import* tools.

Warning

Please note that your existing Setups will not be automatically transferred to newly installed versions of BenMAP. If you wish to preserve your setups, please see instructions for exporting and importing databases in Chapter 9.

4.3 Frequently Asked Questions

I've loaded new baseline incidence rates, but BenMAP-CE won't let me select it in the configuration stage.

When formatting these data for importing to BenMAP-CE, take special care to ensure that you have specified the health endpoints correctly. The baseline incidence rate must be associated with a specific health endpoint and endpoint group in BenMAP-CE. Be sure that you have recorded the endpoint group and endpoint exactly as it is recorded in BenMAP-CE. For example, if the baseline incidence rate is for asthma-related hospital admissions, be sure you have recorded the endpoint group as '*Hospital Admissions, Respiratory*' and the endpoint as '*HA, Asthma*'.

I've loaded a new grid and new population data into BenMAP-CE but I can't seem to use these new data.

Be sure to load the new grid definition first. Next, load the population dataset and be sure to select your new grid definition.

How do I generate a population dataset for a new grid definition?

You can generate a population dataset using a variety of approaches. The key is that you need to have a shapefile of your area of interest (e.g., Census tracts in a city) and you need to have census data matching your area of interest. One source for both a shapefile and the associated population data is the U.S. Census Bureau. (A variety of other agencies have census data, and you need to check around for your area of interest.) Another option for U.S. population data is to use the PopGrid software application, mentioned in Section 4.1.5 and described in Appendix J on Population Data for the U.S. Setup. Using PopGrid, you still need to have a shapefile for your area of interest.

Can I edit a population configuration?

No, you cannot edit a population configuration. You can only view a population configuration. If the population configuration does not match your data, you need to either create a new population configuration to match your data, or reshape your data so that it matches the population configuration.

Chapter 5

Creating Air Quality Surfaces

In this chapter…

- **Define air quality grids**
- **Create air quality surfaces using different methods.**
- **Learn how to structure input files.**
- **Learn how to interpolate monitoring data with Closest Monitor, Voronoi Neighborhood Averaging, or Fixed Radius.**
- **Learn about the Monitor Rollback feature.**

BenMAP-CE is not an air quality model and it cannot simulate changes in air quality. Instead it relies on the model-simulated or monitored air quality inputs given to it. To estimate population exposure to air pollution, BenMAP-CE uses air quality surfaces that it generates from input air quality data (modeling or monitoring data). The program uses a grid structure, where each cell contains a model or monitored air quality value.

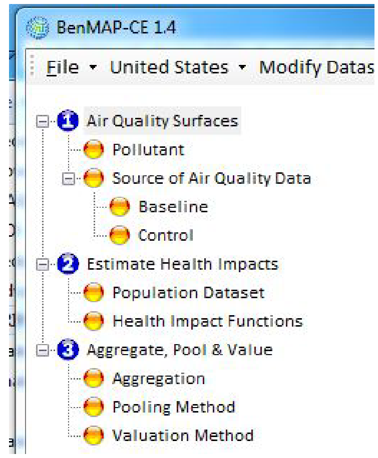
> **Fundamental Concept – Air Quality Surface**
>
> An **air quality surface** contains modeled or monitored air pollution data arranged spatially in a series of cells; these cells may be a regular shape (like a 12km by 12km grid) or an irregular shape (like a county or census tract). BenMAP-CE uses one air quality surface to represent the Baseline scenario and a second surface to represent the Control scenario. These baseline and control surfaces must use the same air quality grid. The program calculates the difference between Baseline and Control surfaces as an input to the health impact functions. Air quality surfaces are stored in files with an .aqgx file extension.

BenMAP-CE creates air quality surfaces to estimate the average exposure to ambient air pollution of people living in a "grid." These grids are either regularly shaped areas like those used by air quality models, or irregular shapes, like provinces, local government areas, cities, or nations. BenMAP-CE does not estimate personal exposure. Instead, the program calculates average pollutant concentration to which people are exposed in each grid cell. BenMAP-CE then uses these average values to calculate health impact functions.

To create air quality surfaces, BenMAP-CE uses a number of inputs, including modeling data or monitoring data. You may enter your own modeling and monitoring data, provided that the data are in a format recognized by BenMAP-CE.

> **Fundamental Concept – Baseline and Control Scenarios**
>
> BenMAP-CE requires both a **Baseline Scenario** air quality surface and **Control Scenario** air quality surface to estimate the effects of a change in air quality (**Delta)**. Both Baseline and Control air quality surfaces are required regardless of whether you use modeling data or monitoring data.
>
> - The **Baseline Scenario** characterizes the air quality levels observed or expected in the absence of the policy change you are evaluating. The baseline is sometimes referred to as "Business as Usual." The baseline scenario is usually considered to be the reference scenario against which to compare a potential scenario characterized by the implementation of regulations.
> - The **Control Scenario** in BenMAP is the scenario in which emissions from one or more source sectors are changed (increased or decreased) from the Baseline scenario. The Control scenario usually represents expected air quality levels after a new regulation or set of regulations has been implemented.
>
> The air quality **Delta** is the change in air pollution between the Baseline air quality grid and the Control air quality grid (Baseline minus Control). BenMAP-CE uses the air quality **Delta** as the input to the health impact function.

To start the grid creation process, locate **Air Quality Surfaces** on the BenMAP-CE tree menu. Under this header, double-click **Pollutant**. On the selection window, click to select a pollutant from the left site and click "Add" button to add it to the right side. You may also click, hold and drag the pollutant of choice from the left side to the right side. Click **OK**.

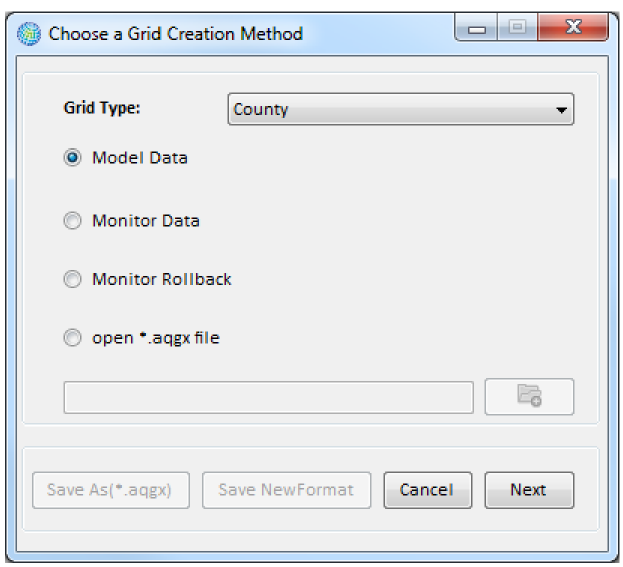


To remove a selected pollutant, click the Remove button or double click the pollutant name to remove it from the left site.

Next, double-click **Baseline** on the BenMAP-CE tree menu to open the **Grid Creation Method** window.

BenMAP-CE will then ask which **Grid Type** (which you select from a previously loaded shapefile) to use and which of the following types of air quality data you wish to use:

- *Model Data*. Choose this option if you have air quality modeling data that you wish to use directly. Table 5-1 below describes the input format for modeling data.
- *Monitor Data*. Choose this option if you wish to use point source monitoring data (measured observations).
- *Monitor Rollback*. Choose this option if you want to reduce monitor levels by a specified amount.
- *Open *.aqgx file.* Choose this option to import a file that has already been created.



Select your **Grid Type** and then click **Next.** BenMAP-CE will direct you through the necessary steps for each option (described below).